

Software Engineering

Declarative prototyping - Exercises -

Exercise 1

- Quicksort with difference lists – declarative prototype:

```
qsd :: [Int] -> [Int] -> [Int]
qsd []      ys = ys
qsd (x : xs) ys =
    let (ls,gs) = partit x xs
    in qsd ls (x : qsd gs ys)
```

```
partit :: Int -> [Int] -> ([Int],[Int])
partit p []      = ([],[ ])
partit p (x:xs) =
    let (ls,gs) = partit p xs
    in  if (x < p) then (x : ls,gs) else (ls,x:gs)
```

```
Main> qsd [1,7,3,12,4,2,5] [0,0,0]
[1,2,3,4,5,7,12,0,0,0]
```